

2.1: Intro to Data Visualization

1. What question(s) or project goal(s) relate to this spatial aspect? Did you address any of them in your previous analysis?

The key project goals related to the spatial aspect include:

- Determining the differences in staffing needs across each state.
- Identifying geographic trends in influenza-related deaths and vaccination rates.
- Understanding how influenza peaks vary geographically and temporally.

In the previous analysis, we addressed some spatial aspects by analyzing the correlation between the population of older individuals and influenza-related deaths across states.

2. List at least one way in which visualizing the data by states might help.

Visualizing the data by states helps identify regional patterns and trends, such as areas with higher influenza-related mortality or vaccination rates. It also allows for easy comparison of staffing needs across states, highlighting where additional resources may be required.

3. What would you be visualizing (or comparing) across each state?

We would be comparing:

- The number of influenza-related deaths.
- Vaccination rates and their impact on mortality.
- Peak influenza activity periods.
- The proportion of vulnerable populations, such as older adults or children, receiving flu shots.

4. Identify any other questions you had from the previous achievement that weren't answered by your analysis and discuss how visualizations may assist you in answering them.

Several questions from the previous achievement were not fully addressed, and visualizations can assist in answering them:

- **Does influenza peak in every state during the same month?** To show this visually, we could use a heat map or a time-series graph that highlights peak influenza activity periods across different states.
- **Do certain states have more vulnerable populations?** A choropleth map showing the density of older adults or other high-risk groups can help visualize this.
- **Do states vary in how many kids receive the flu shot?** Spatially showing different population proportions can be achieved with maps displaying vaccination rates among children across states.

These visualizations would provide a clearer understanding of geographic trends and help in strategic planning for staffing and resource allocation during influenza seasons.